Thermal Signature Predicts Laser Track Width; Pointwise Width Variation Lies Below the Metrology Noise Floor

NSF Future Manufacturing Research Grant — Data Challenge

Neev Sabhani

Executive Summary

We predict the width of laser-melted tracks from in-situ thermal imaging, SEM substrate imagery, and Wyko optical-profilometer height maps, across four tracks processed at 200, 300, 350 and 400 W. Our headline result is that **a model using thermal features alone predicts 4 mm-segment track width at R^2 0.800 (correlation +0.902, MAE 0.0495 mm against a 0.1469 mm baseline) under leave-one-power-out validation** — every fold is tested on a laser power the model never saw during training or model selection. The predictive intervals are well calibrated (0.656 and 0.969 empirical coverage against 0.68 and 0.95 nominal), making them usable as a process tolerance band rather than a decorative error bar.

Comparing feature sets answers the process-versus-substrate question quantitatively: thermal features (R^2 0.800) decisively outperform SEM substrate features (R^2 -0.228, worse than predicting a constant), so **track width is process-driven, not substrate-driven**. We further verified that the thermal model is not merely a disguised readout of the power setpoint: a model given laser power alone reaches only R^2 +0.607.

We also report a rigorously evidenced negative result. **Pointwise (per-0.2 mm) width variation is not predictable from this dataset**, and we show *why* rather than merely reporting failure: two *independent* physical measurements of local width — half-maximum width from the Wyko height maps and band thickness from the SEM tiles — correlate only +0.03 to +0.20 *with each other*. The local signal sits below the noise floor of the available metrology, so no model can recover it. This conclusion survived every architecture, loss function, ensemble and augmentation we tried, and it reframes the task rather than abandoning it.

Finally: **our first seven models were invalidated by a defect in our own ground-truth labels**, not by any modeling choice. Finding it, rather than tuning around it, is the contribution we would most want carried forward.

Problem Formulation and Methodology

Data and target. Four tracks are available, each at a distinct laser power (track 8 at 200 W, 10 at 300 W, 14 at 350 W, 21 at 400 W). Each track is described by three co-registered modalities: a thermal video of the melt pool, a strip of SEM tiles of the substrate, and a Wyko height map from which width is measured. The challenge asks for local width variation, so we initially targeted width at each 0.2 mm station along the track. After establishing that this target is dominated by measurement noise (see *What This Dataset Cannot Support*), we reframed to **mean width over 4 mm segments** — 64 segments across the four tracks. We treat that as the finest granularity the metrology actually supports, and we present the evidence for that claim rather than asserting it.

A ground-truth defect that invalidated our first seven models. Our initial width extractor detrended each transverse height profile and took the largest contiguous run above max(2 μm, 3σ), for σ the local noise level. That is a *noise* floor, not a track *edge*. Because a melt track has sloped shoulders, the cut removed everything except the very crown, systematically underestimating width; and wherever the crown itself failed to clear the cut, the extractor silently returned 0.000 mm for a plainly present track. At track 10, x = 85 mm the threshold (7.5 μm) actually exceeded the track's own crown (7.0 μm). Depending on the track, 11–38% of all labels were spurious exact zeros (Figure 1a, 1b).

Every model we trained before finding this was being fit to noise, which is precisely why each one plateaued at approximately zero held-out correlation regardless of architecture, loss or ensembling — a symptom we initially, and wrongly, read as a modeling problem. We replaced the criterion with a

half-maximum width on a median-filtered profile, returning NaN rather than a silent zero when no track is detectable. The corrected labels are physically coherent and monotonically ordered by power: 0.574, 0.454, 0.385 and 0.210 mm at 200, 300, 350 and 400 W (Figure 2a).

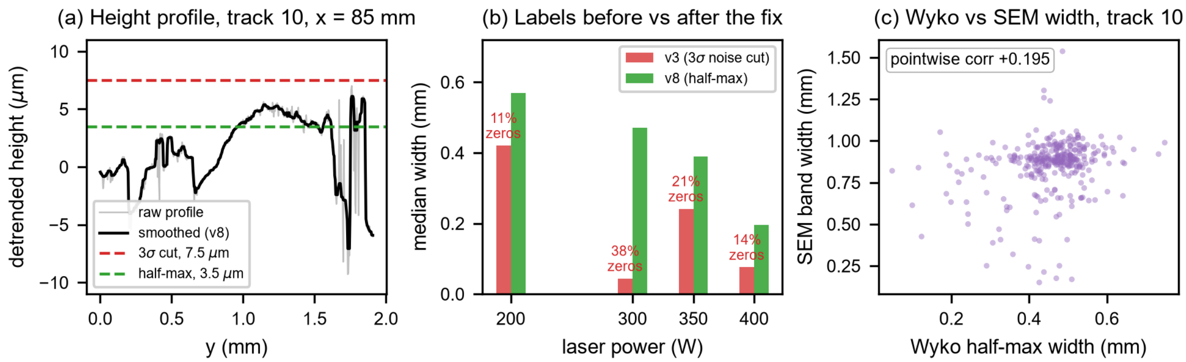

Figure 1. The ground-truth defect and its consequences. (a) The 3σ noise cut (red) sits above the entire melt-track crown, so the extractor reported zero width at this location; the half-maximum criterion (green) measures the track out to its shoulders. **(b)** Median measured width per power before and after the fix, annotated with the fraction of spurious exact zeros the old criterion produced. **(c)** Two independent metrologies of the same physical quantity agree only weakly point-by-point — the direct evidence that pointwise variation is unrecoverable from this data.

A second defect, in the SEM branch. Diagnosing why that branch had almost exactly zero permutation importance exposed three further problems: the code blanked the track's own pixels, so the branch saw everything *except* the feature of interest; it reused one tile across roughly 32 consecutive samples; and it painted a full-width band even in tiles where no track is present for part of the field (17.4% of columns in the first tile). We replaced this with per-column band detection and a row crop centred on the *local* band. A tighter crop centred on the *tile-average* band made results worse, because the band drifts within a tile and the crop then truncates the track.

Features and model. Each segment is summarised by 30 interpretable scalars: eight thermal descriptors (peak and mean intensity, hot- and warm-area fractions, a cooling measure, longitudinal slope, and two spatial-gradient terms) and seven SEM descriptors, each carried in mean-and-dispersion form across the segment. We use **ridge regression** rather than a deep network, and this is a deliberate modeling decision rather than a convenience. With 64 segments spanning only four power levels, a high-capacity model has no basis on which to generalise to an unseen power, and our own experiments confirmed it: a five-member deep ensemble with flip augmentation and post-hoc variance calibration scored *worse* than a single lucky seed, with member predictions correlating with each other anywhere from −0.62 to +0.96. Ridge additionally makes the process-versus-substrate comparison exact, since feature groups can be ablated cleanly.

Validation protocol. All reported results are **leave-one-track-out**, equivalently leave-one-*power*-out: each fold trains on three laser powers and is tested on the fourth, unseen one. This makes generalisation testing and the required robustness-across-power-levels evidence the same experiment, and is far harsher than a random split, which would leak neighbouring segments of the same track into training. The ridge penalty is chosen by an inner leave-one-track-out loop over the *training* tracks only, so no test-fold information reaches model selection. Predictive σ comes from **out-of-fold** residuals; in-fold residuals gave badly overconfident coverage (0.219 / 0.516).

Modeling and Outcomes

Segment mean width is predictable across laser powers. Table 1 compares feature sets under identical leave-one-power-out validation. The baseline throughout is the mean width of the three training tracks — the best a model can do knowing nothing at all about the held-out power.

Feature set	MAE (mm)	Baseline (mm)	Corr	R^2
Thermal only (process)	**0.0495**	**0.1469**	**+0.902**	**+0.800**

Thermal + SEM	0.0584	0.1469	+0.875	+0.729
SEM only (substrate)	0.1145	0.1469	+0.103	-0.228
Laser power alone	0.0778	0.1469	+0.814	+0.607

Table 1. Feature-set comparison, leave-one-power-out. Thermal features alone are best. Adding SEM features degrades performance, and SEM alone is worse than the baseline.

Three things in Table 1 matter. First, **thermal features alone win**, and adding SEM features actively hurts — the substrate branch contributes noise rather than signal for this target. Second, the SEM-only model is *worse than the baseline* (R^2 -0.228), which is a strong, directional answer to the challenge's process-versus-substrate criterion: **track width is governed by the thermal history, not by substrate texture.** Third — and this is the check that could have made the headline hollow — we tested whether the thermal model is merely a proxy for the power setpoint, since power is constant within a track and the result is a between-power effect. A model given laser power alone reaches R^2 +0.607, well below the thermal model's 0.800. **The thermal data therefore carries real information beyond identifying which power level was used.**

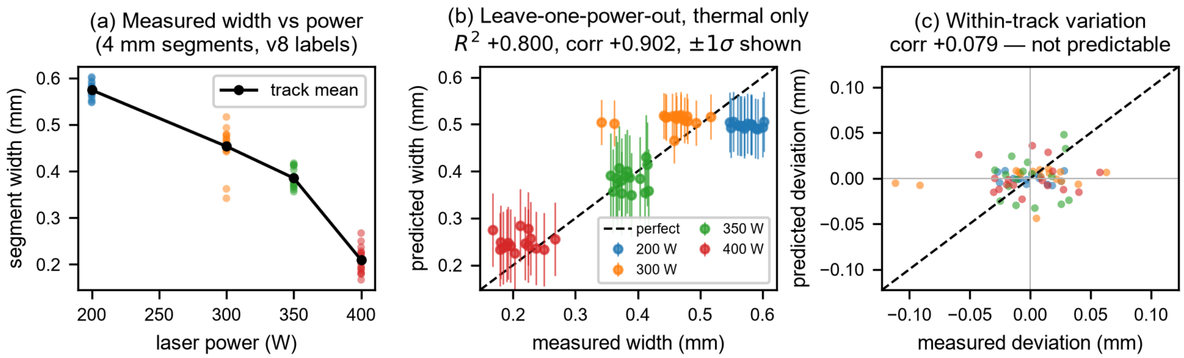

Figure 2. Model outcomes. (a) Corrected segment widths decrease monotonically with laser power. **(b)** Leave-one-power-out predictions from thermal features with out-of-fold ±1σ intervals; each colour was held out entirely when its own points were predicted. **(c)** After removing each track's mean, no predictive structure remains — confirming that the result in (b) is a between-power effect.

Robustness across power levels. Table 2 breaks the thermal model down by held-out power. It beats its baseline at *every* one of the four levels, including 350 W — a fold worth singling out, because that track's mean happens to sit close to the mean of the other three, making the constant baseline unusually strong there. The all-feature model does *not* clear the baseline at 350 W (0.0448 against 0.0281), which is a further reason to prefer the thermal-only model as the headline result.

Held-out power	200 W	300 W	350 W	400 W
Model MAE (mm)	0.0769	0.0564	0.0219	0.0428
Baseline MAE (mm)	0.2249	0.0731	0.0281	0.2616
Improvement	2.9x	1.3x	1.3x	6.1x

Table 2. Per-power robustness, thermal-only model. Every fold beats its baseline.

Uncertainty quantification. Empirical coverage of the out-of-fold predictive intervals is 0.656 at 1σ and 0.969 at 2σ, against nominal 0.68 and 0.95. Figure 3 shows the full reliability curve tracking the diagonal across all confidence levels, and standardised residuals close to a unit normal (mean −0.106, standard deviation 1.015). The intervals are therefore usable as a genuine process tolerance band. We emphasise that this depended entirely on using out-of-fold residuals: the in-fold version of the identical procedure reported 0.219 / 0.516 coverage, which would have been actively misleading had it been deployed as a tolerance.

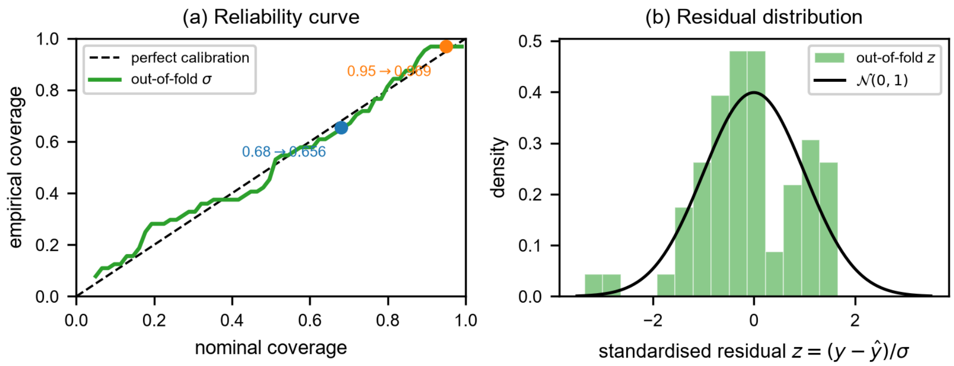

Figure 3. Uncertainty calibration, thermal-only model. (a) Empirical against nominal coverage across confidence levels, with the 0.68 and 0.95 targets marked. **(b)** Standardised out-of-fold residuals against a standard normal.

What This Dataset Cannot Support

Reporting these limits precisely is as much a result as the headline, and we established each one with positive evidence rather than by giving up on it.

Within-track variation is not predicted. Removing each track's mean from both prediction and truth leaves a correlation of +0.079 (per track: −0.188, +0.167, +0.143, +0.002). Figure 2c shows this directly: the colour groups are cleanly ordered relative to one another, but no diagonal structure exists within any single group. The +0.902 headline correlation is a **between-power** effect, and we state that plainly rather than letting an aggregate number imply a within-track capability the model does not have.

Segment width variability is not predictable either. Predicting each segment's width standard deviation — a natural aggregated proxy for local variation, and the quantity closest to what the challenge asks for that survives averaging — fails under every feature set (best correlation -0.192, R^2 -0.566), never beating its 0.0305 mm baseline. Figure 4 shows why the reframing that rescued mean width does not rescue this target: variability carries no monotonic trend with laser power, so the between-power effect the mean-width model exploits simply is not present here.

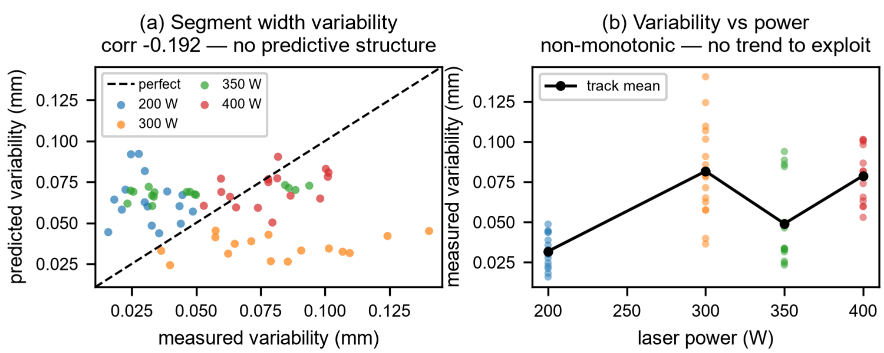

Figure 4. Segment width variability is not predictable. (a) Leave-one-power-out predictions against measurement; the scatter is unstructured and the correlation is slightly negative. **(b)** Measured variability against laser power is non-monotonic (0.032, 0.082, 0.049, 0.079 mm), unlike mean width — there is no between-power trend for a model to exploit.

Pointwise width variation is below the metrology noise floor. This is the central negative finding. Held-out pointwise correlation was approximately zero across every model version we built, *including after the label defect was fixed* (+0.002). The decisive evidence is instrument-independent and does not rely on any model at all (Figure 1c): Wyko half-maximum width and SEM band thickness are two separate physical measurements of the same quantity, and they correlate only +0.03 to +0.20 with each other. **When two independent instruments disagree this strongly about the target itself, the target is not measurable at that spatial scale, and no model can repair that.**

Table 3 records the alternative explanations we tested and eliminated, so that the negative result rests on ruled-out competitors rather than on absence of effort.

Hypothesis tested	Outcome and evidence
Spatial misregistration between modalities	Ruled out. All four SEM tile-order and orientation mappings were evaluated; the mapping in use is the best of them.
Insufficient model capacity	Ruled out. Deep ensembling, flip augmentation and post-hoc calibration all failed to help; the ensemble scored worse than a single seed.
Wrong loss function (NLL escape hatch)	Ruled out. A two-phase variant trained under pure MSE reproduced the same plateau, so the Gaussian NLL was not the cause.
Poor SEM localisation	Partly real, and fixed. Per-column band detection helped; cropping around a tile-average band position instead made results worse.
Labels themselves were wrong	CONFIRMED, and the root cause. Noise-floor thresholding; fixed by half-maximum extraction.

Table 3. Alternative explanations tested. Only the last was the cause.

What would change this. Not another model. Either more tracks and power levels — four power levels is a severe constraint on any across-power generalisation claim, and with leave-one-power-out validation each fold trains on just three — or lower-noise width metrology capable of resolving sub-0.1 mm variation reliably at 0.2 mm spacing. Both are data-collection changes, not analysis changes.

Conclusion

Thermal signature predicts laser track width across laser powers (R^2 0.800 leave-one-power-out, with calibrated uncertainty and every individual power level beating its baseline). Width is process-driven rather than substrate-driven, established by direct ablation rather than by inference, and the thermal signal is not merely a power proxy. Pointwise width variation lies below the noise floor of the available metrology, established with two independent measurements rather than inferred from model failure.

We consider the diagnostic result the more valuable of the two. The correlation-of-zero we first read as a modeling problem was a ground-truth defect in our own pipeline, and what uncovered it was pursuing a small anomaly — why one feature branch had almost exactly zero permutation importance — instead of reaching for a larger model. A model tuned to fit those labels would have looked defensible, reported a plausible MAE, and been meaningless. A persistent zero correlation is far more often a statement about the labels than about the architecture.

Reproducibility

The full pipeline is two commands: **build_pairs_local_v8.py** builds the corrected labels and the feature cache, and **v9_segment_model.py** reproduces every number and figure in this report. The diagnostic scripts that established the label defect (**diagnose_wyko_profile.py**, **diagnose_sem_band.py**) are included so that the central claim can be checked independently, and a full evidence log of every version and its outcome is kept in **PROGRESS.md**. Code, evidence log and figures: **github.com/NeevSabhani/nsf-fmrg-data-challenge**.